



A smart anomaly-based intrusion detection system for the Internet of Things (IoT) network using GWO–PSO–RF model

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Abstract

The Internet of Things (IoT) is adding the advancement in the technology for creating smart environments to facilitate humans for various works. The technological developments provide many comfort and opportunities to the businesses and open the doors for the intruders or attackers to explore and exploit various attacks to evade the IoT networks' security. Hence, security and privacy are the key anxiety to the IoT network model. Protection of computer and IoT networks from various types of attacks and threats is necessary. The traditional intrusion detection system (IDS) collects and uses massive data with unnecessary, irrelevant, and inappropriate features, which cause high detection time and low accuracy. This paper proposes an IDS to identify various attacks for IoT networks. A combination of Grey Wolf Optimization (GWO) and Particle Swarm Optimization (PSO) is used to extract relevant IoT network features. The extracted features are fed to a random forest (RF) classifier to achieve high attack detection accuracy. The experiments are conducted in the python programming environment to evaluate the proposed model on KDDCup99, NSL–KDD, and CICIDS-2017 datasets. The proposed GWO–PSO–RF NIDS model has achieved an average accuracy of 99.66% for multiclass classification. The accuracy of the proposed model has been compared with other similar approaches to show its effectiveness. The work presented here also addresses the issue of data imbalance.

Keywords IDS · IoT · Random forest · Grey wolf optimization (GWO) · Particle swarm optimization (PSO) · Feature selection (FS)

1 Introduction

Due to advancements in technology such as IoT, Cloud Computing, the network devices and services are increasing continuously, increasing the complexity of network that causes the challenges in maintaining network security due to increasing complexity of network. International Data Corporation (IDC) reported that 41.6 billion IoT devices would be connected by 2025 [1]. The development of connecting worlds to the consumers is giving a lot of opportunities to the businesses. It also invited intruders to explore and exploit different types of attacks to evade IoT network security. The security attacks in a network may cause a massive loss to the companies and consumers in terms of money and reputation. Hence, it is essential to take care of the safety of the networks. The networks may belong to IoT or others. Various tools and solutions are there to fight with different types of network attacks, such as firewalls, antimalware, and spam filters. Examples of various tools and techniques are IDS, Digital forensics, access control,

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auditing, log analysis, Unified Threat Management (UTM), Intrusion Prevention System (IPS), and management of risk [2]. However, the intrusion detection system (IDS) can be an essential and highly valuable security tool to ensure the security of IoT network. An IDS detects the network traffic attacks and sends alerts to the network administrator by mentioning source IP, destination IP, and types of attacks.

The IDS is placed outside of network infrastructure and performs the analysis on a copy of the inline network traffic. Therefore, real inline network performance is not affected. Based on IDS placement, it is of types: HIDS—Host-based IDS and NIDS—Network-based IDS. HIDS collects data and detects intrusions on a particular machine, known as the host. A sensor is installed on each host, known as host-agent, to collect data of specified events. Network-based IDS (NIDS) scans and audits network traffics at the router or host level generate alert for malicious or suspicious packets. In this modern scenario of growing Internet, network-based IDS (NIDS) is very suitable.

The NIDS may be a hardware or software-based system attached to a network medium such as FDDI and Ethernet. The other categories of NIDS are—signature-based NIDS, anomaly-based NIDS, and hybrid NIDS. Various approaches are there to identify intrusions such as machine-learning (ML) techniques, deep-learning (DL) techniques, feature engineering, and ensemble techniques. The NIDS can detect attacks from various systems and networks by analyzing the collected information [3]. The network intrusion detection system (NIDS) is a widely used essential technique for ensuring network security. It is a dynamic defensive approach for detecting and responding to attacks, intrusions, and malware effectively. A NIDS works as a tool to sniff, monitor, store, and analyze users, services, and networks to manage networks and identify various security vulnerabilities timely [4]. A network-based intrusion detection system (NIDS) works in three phases. In the first phase, it monitors the packets received from network-based sensors or host-based sensors. The second phase tries to extract features with feature extraction or feature selection. In the last step, it detects the intrusion or anomaly-based on classification techniques from the extracted features.

The NIDS works on the captured copy of network traffic to see the harmful activities [5]. Hybrid techniques are in the trend to design the IDS since it is adaptive and can handle complex logic without modifying the existing systems. The hybrid techniques agents are trained with classification rules and used cluster classification (anomaly-based IDS). Most of the already existing NIDS uses all attributes of the dataset generated by the network traffic. However, some of the attributes are irrelevant and not required in the intrusion detection process. Hence, feature optimization is necessary to decrease the detection time, false alarm rate (FAR) and increases the detection rate

(DR). The scope of ML and DL techniques to deal with anomaly detection is suggested in [6]. Several approaches are proposed by the many researchers for identifying and classifying attacks, such as genetic algorithm and fuzzy logic algorithms [7], clustering and K-nearest neighbor (k-NN) algorithms [8], Decision Tree and Random Forest [9], SVM, and RBM together [10], and convolution neural network (CNN) [11]. Advantages of the ML-based IDS model are as follows:

1. Attack with small variation can be detected easily since the ML-based models are trained on the pattern/behavior basis of network flow and might work for most scenarios.
2. Zero-day attacks can easily be detected if the model is trained through unsupervised learning algorithms.
3. ML-based IDS are faster as well as higher detection accuracy, even in the complex network environment.

There are challenges in applying the ML to IoT Network given below as:

1. *Low processing capacity* IoT devices are small in size and have limited processing capacity with energy constraints. ML algorithms require processing the real-time data that create challenges in the implementation of such resource-constrained environments.
2. *Data analytics* In the IoT environment, data are generated heterogeneously, which requires preprocessing to be applied in ML. It demands the processing power and memory space in the IoT devices. Hence, it creates a challenge for providing an efficient solution for diversified data.

The main focus of this paper is to use a hybrid GWO–PSO for feature selection (FS) to decrease processing cost, memory and apply a Random Forest (RF) for fast and efficient classification from the selected features. The considered parameters for the performance analysis are accuracy, Precision, recall, and F1-score. The proposed NIDS model's deployment is regarded outside of the IoT network and before the router on a device to address the mentioned challenges, as presented in Fig. 1. It is believed that the device has sufficient processing power and memory to support the proposed ML-based NIDS model in IoT Network. Further, the proposed ML-based NIDS has been evaluated on different datasets such as KDDCup99, NSL–KDD, and CIC–IDS-2017 to manage the heterogeneity.

1.1 Contribution

The contributions are as follows:

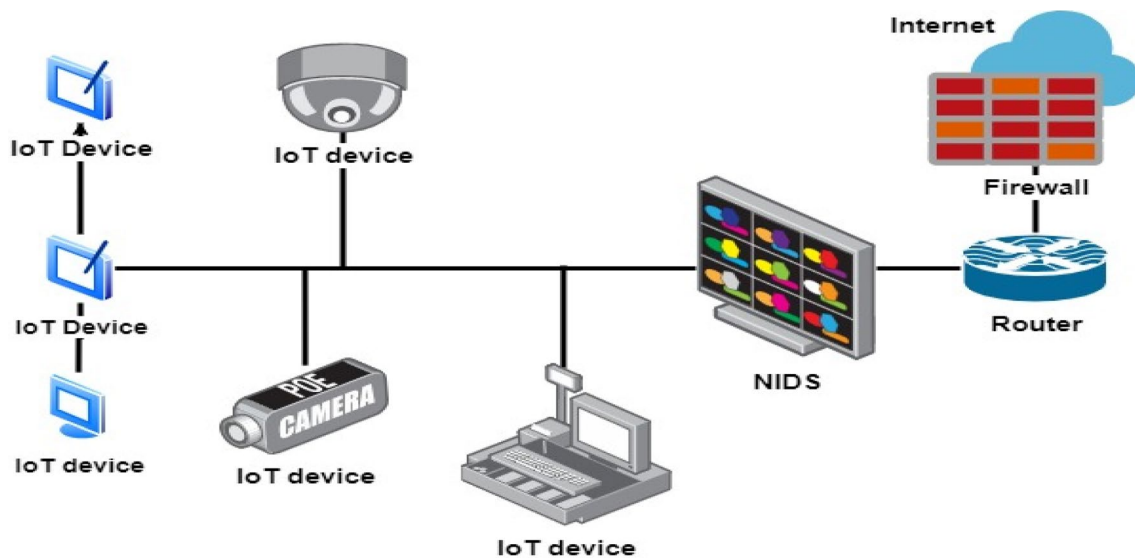


Fig. 1 General deployment of a NIDS

1. An efficient intrusion detection framework has been proposed to discover intruders from the IoT network more effectively where a knowledge base is added to detect the unknown attacks.
2. For effective intrusion detection, the highly supported features are obtained by the Hybrid GWO–PSO. Hybrid GWO–PSO is the first time developed for IoT network intrusion detection, which follows a significant fitness function to remove redundant and irrelevant features.
3. Oversampling has been used to address the issue of imbalanced data.
4. For anomaly-based intrusion detection, a group of popular ML algorithms is compared in classification from selected feasible features. It has been observed that Random Forest (RF) with hybrid GWO–PSO has achieved best performance among the consider ML algorithms. Hence, the GWO–PSO–RF NIDS model has been proposed for the anomaly IoT network.
5. The obtained results are equated with (i) existing ML strategies and (ii) most representative existing related works on KDDCup99, NSL–KDD, and CICIDS-2017 datasets.

1.2 Organization of the paper

The remaining paper organizations are as follows.

Section 2 deals with the related work. Section 3 discusses the proposed framework for the IoT network. Section 4 elaborates on the developed NIDS based on the proposed framework. Section 5 deals with the implementation and evaluation of the proposed GWO–PSO–RF NIDS model for

the IoT network. Finally, Sect. 6 provides the overall conclusion about the paper with the future direction of work.

2 Related work

In recent years, researchers have attempted to resolve the issues of intrusion detection in the IoT network. As mentioned earlier, these researches are carried out using various techniques such as ML, semi-supervised learning, adaptive, heuristic, decision tree, and DL. This section presents the related works on handling unbalanced data issues in network-based intrusion detection and other recent intrusion detection proposals for IoT networking environments. At last, they are summarized in Table 1 comparatively.

Rodda [12] has designed a NIDS for multiclass classification by utilizing the multi-layer perceptron (MLP) and radial basis function (RBF). However, the proposed model results were excellent but unable to handle minority classes such as R2L and U2R. Data imbalance in different applications is discussed in [13], where the authors find only two main methods to handle this issue: data-level and algorithm-level. Data-level preprocessing techniques balance the ratio of various classes. Examples of data-level techniques are oversampling, under-sampling, and SMOTE variations. At the algorithm-level technique, fine-tuning is used in the classifier to increase detection capability for minority classes. Abdulhammed et al. [14] have applied the four balancing techniques on the CIDDs-001 dataset: down-sampling, Upsampling, Spread sub-sample, and Class Balancer. RF performed well in comparison to DNN, Variational AE, Voting, and Stacking algorithms.

Table 1 Summarization of the recent existing proposal and the proposed approach

References	Method to handle data imbalance issue	FS technique	Classification technique(s)	Attack model for Prediction	Dataset(s) used
Rodda [12]	Not used	Not used	Multi-layer perceptron and radial-basis function	Network Intrusion Detection System (NIDS)	NSL–KDD
Abdulhammed et al. [14]	Down-sampling, Upsampling, spread sub-sample and Class Balancer	Not used	Variational AE, DNN, RF, Voting, and Stacking	Network Intrusion Detection System (NIDS)	CIDDS-001
Xiao and Xiao [15]	Random oversampling	Not used	Simplified Residual Networks	Network Intrusion Detection System (NIDS)	NSL–KDD
Hamid et al. [16]	SMOTE	Principal Component Analysis (PCA)	Support Vector Machine (SVM)	NIDS for binary classification	KDD99
Dhaliwal et al. [18]	Not used	Not used	XGBoost	NIDS for binary classification	NSL–KDD
Verma et al. [19]	Not used	Not used	XGBoost and Ada-Boost	NIDS for multi-classification	NSL–KDD
Kaja et al. [20]	Not used	Not used	K-means, J47, Random Forest, Naive Bayes, and Adaptive Boosting	NIDS for binary classification	KDD99
Lee and Park (2019) [21]	Generative Adversarial Networks (GAN)	Not used	RF	NIDS for multi-classification	CICIDS-2017
Tkachenko et al. [22]	Not used	Not used	GRNN–SGTM	Recovering lost data	Self-collected IoT-based dataset
Venkatraman and Surendiran [27]	Not used	Not used	Own developed algorithms	Signature-based, anomaly-based, and hybrid-based IDS to detect DoS attacks	Own developed dataset
Yanmiao et al. [31]	Not used	Not used	Multi-CNN fusion model	NIDS for binary classification and multi-class classification	NSL–KDD
Mohamed et al. [32]	Three RDTIDS model	Not used	Rules and Decision Tree	NIDS for binary classification and multi-class classification	CICIDS-2017 and BoT-IoT
Miloud et al. [38]	Not used	Not used	J48, Byes Net, Random Forest, and Hoeffding Trees	Network Anomaly Detection for multi-class classification	NSL–KDD
Torres et al. [34]	Not used	Not used	Recurrent neural networks	NIDS for multiclass classification	NSL–KDD
Keserwani et al. [37]	Not used	GWO–CSA	DSAE	NIDS for binary classification and multi-class classification	CICIDS-2017, UNSW-NB-15, NSL–KDD
Proposed approach	Oversampling	GWO–PSO	Random Forest, DT, LR	NIDS for multiclass classification	KDDCup99, NSL–KDD, CICIDS-2017

Xiao and Xiao [15] have utilized Simplified Residual Networks to create a NIDS and test on the NSL–KDD dataset. In the experiment, random oversampling balanced the dataset. This technique duplicated the minority class samples and caused the overfitting of data. Hamid et al. [16] have used the SMOTE concept of [17] to handle the class imbalance problem. The authors performed FS on the KDD99 dataset, and SVM was used as a classifier

for selected features, but the detail was missing to detect minority class attacks.

Recently, different tree-based ensemble methods are used in network intrusion detection. Dhaliwal et al. [18] have applied XGBoost in their proposed NIDS, and the NSL–KDD dataset was used to evaluate performance. Further, in [19], the authors have used the amalgamation of XGBoost and Adaboost for the separation of benign and

malicious traffic. Kaja et al. [20] have followed a comparative analysis of DT, RF, Naive Bayes, J48, and Adaboost methods for separating the attacks recognized by k-Means clustering. It was observed that the performance of RF was best. However, the class-wise assessment was absent in the results. Lee and Park [21] have tackled the class imbalance problem of the CICIDS-2017 dataset through an unsupervised technique known as GANs and generated similar samples for existing minority classes. The RF was used as a classifier. Tkachenko et al. [22] have utilized the combination of General Regression Neural Network (GRNN) and Successive Geometric Transformation Model (SGTM) for improving the prediction accuracy for recovery of missing IoT data. Experiments on the public datasets were missing for performance comparison.

Many researchers focused their research on for DoS attack detection in IoT network environment such as [23–26]. In addition, many more researchers have presented their proposal for network intrusions in the IoT environment. Miloud et al. [28] proposed a security framework based on ML to mitigate various vulnerabilities caused by various threats. This framework presented the combinations of monitoring agent and reaction agent, where the combination utilizes ML techniques for identifying anomaly-based in IoT systems. In particular, the distribution of attacks successfully found and achieved an accuracy of 99.71%. Venkatraman and Surendiran [27] have proposed hybrid IDS utilizing the analyzed files related to multimedia. They used online collections as repositories to train the IDS model and identify the intruder in IoT networks. The obtained experimental results were accurate up to 99.06% for identifying different attacks such as DoS, control hijacking, a new one, and so on in IoT environments. Man et al. [28] have presented the IDS model for the IoT environment, where they used ULEACH clustering approach to improve node utilization to enhance the performance. They also used PSO for balancing intrusion identification efficiency. Mohamed et al. [29] have detailed the latest IDSs for the IoT environment with their methods, mechanisms, and features. Yahya and Farookh [30] have investigated various datasets for utilizing them in IoT environments to test various techniques for making a smart environment. After that, they presented a data collection framework in real time for intrusion detection so that IDS for the IoT environment can be evaluated and tested. Yanmiao et al. [31] proposed an IDS for IoT environment where they used multi-CNN, a DL technique, and a fusion method in their IDS model for identifying intrusions. Multi-CNN visualized the intrusion detection problem, and fusion classified the intrusions. They achieved high accuracy on the NSL-KDD dataset for binary classification and multiclass classification. Mohamed et al. [32] have proposed a novel IDS model for IoT network environments, namely RDTIDS. Then, the model used the combinations of various

classifiers based on DT and rules such as JRip, Forest, and REP Tree. They evaluated their model in on CICIDS-2017 and BoT-IoT datasets. Jagadeesh and Raji [33] explored and proposed a specification heuristic approach to assess intrusion's scope in the IoT network. Torres et al. [34] analyzed the recurrent neural network (RNN) for detecting the traffic behaviors and observed in some of the cases the RNN model is not performing well on traffic behaviors as well as for imbalanced network traffic. For conducting experiments, they used transformed data as a sequence of characters to the RNN. Yin et al. [35] proposed the RNN-IDS model to perform the classification of attacks on the NSL-KDD dataset instead of pre-training. The results of the proposed model were compared with other ML-based classifiers results such as RF, SVM, and J48 for binary as well as multiclass outputs. They showed that the RNN-IDS model identified the intrusion well with the high accuracy. Mehta et al. [36] has presented an IDS for IoT environment which uses the relationship among sensor's time series data using AI-based techniques for detecting the anomalies. Keserwani et al. [37] have proposed a GWO-CSA-DSAE NIDS model for virtualized cloud network (VCN), where GWO-CSA was used for feature selection and deep sparse auto-encoder (DSAE) was used as classifier. Table 1 summarizes the related works in comparative manner.

However, many existing IDSs contributed well to attack detections, but still, some limitations are there:

1. In the data analysis, all raw data are considered and required to analyze several times. It works as an extra burden on cyber-security systems. Hence, it caused insufficiency in the system.
2. Since data diversity is there in the real environment and existing IDSs cannot perform on low-dimensional data and high-dimensional data.
3. Most of the existing IDSs use the higher computation in the packet inspection for detecting the attacks [39].
4. High dimensional data can contain some irrelevant attributes and leads to high computing cost and increased execution time.
5. Addressing data unbalancing issues and selecting relevant features to support better classification is still a challenge.

The survey it was found that random forest performance is better than other ML-based approaches such AdaBoost algorithm, XGBoost, or the GRNN-SGTM ensemble to identify the intrusions from the selected features. Further, RF takes less time to produce the result, and, most important, it is simple to understand and implement. Our proposed intrusion detection framework is well suited for the imbalanced and high-dimensional IoT network environment for detecting the intrusions or anomaly. Hence, the works are

towards the anomaly or intrusion detection such as DoS attack and DDoS attack in the IoT network.

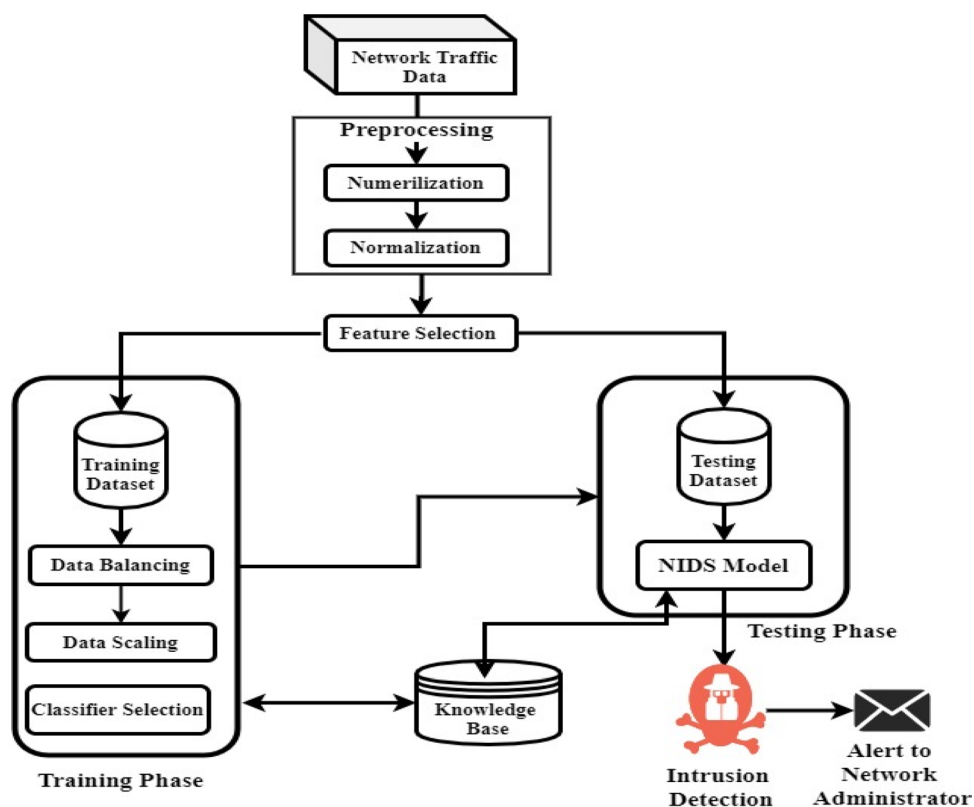
3 Proposed methodology

An efficient intrusion detection framework has been proposed to detect intrusions in the IoT network environment effectively. The approach of the proposed framework with its sub-modules has been depicted in Fig. 2.

There are seven modules used in the approach of the proposed framework given as the following:

1. **Data collection module**
 2. **Data preprocessing module**
 3. **Feature selection module**
 4. **Training module**
 5. **Knowledgebase module**
 6. **Testing module**
 7. **Alert module**
1. **Data collection module** In this module, the dataset can be prepared from the network flow using different logging tools so that a model can be trained to identify the intrusion. Network-related publicly available datasets such as KDDCup99, NSL-KDD, and CICIDS-2017 can also be considered for the data collection.
 2. **Data preprocessing** Real-life data are in different forms, such as structured, unstructured, videos, images, and audio. In this module, data are transformed or encoded, cleaned, normalized, balanced using numerilization, and normalization to be parsed through the machine in the ML process easily.
 3. **Feature selection module** Modern datasets contain vibrant information collected from many IoT devices and sensors, making them high-dimensional data. This high-dimensional data include some of the irrelevant features, harm the model performance. The feature selection module helps select relevant feature subset (automatically or manually) to avoid or decrease overfitting, improves accuracy, reduces training time, reduces data size, and improves performance.
 4. **Training module** In this module, the data belong to selected features that are balanced using a data-balancing method. The balanced data work as an input to the selected classifier. The classifier chosen may be an ML-based classifier or a DL-based classifier.
 5. **Knowledgebase module** This module contains a set of rules like “if-then” statements, for induction. It gets updated and updated by training module and model so that the new attacks can be identified. It provides decisions using facts or knowledge as well as rules.
 6. **Testing module** The testing module uses the testing data on the model to evaluate the model performance

Fig. 2 Proposed framework for intrusion detection in the IoT network



for detecting intrusion in the IoT network environment based on various parameters such as accuracy, Precision, recall or detection rate (DR), F1-score, and false alarm rate (FAR).

7. *Alert module* On detecting the intrusion or attack, an alert is sent to the network administrator. The alert contains the victim IP, attacker's IP, detection result, destination port, and source port. The alerted intrusions are stored in the central log.

4 IDS model proposal based on the proposed framework

In this section, a GWO–PSO–RF IDS model for IoT network has been proposed based on the IoT network's proposed framework. The details of the proposed model are as follows:

4.1 Dataset collection

Before implementing a NIDS in the real world, it is required to train it on the data samples to learn real network traffic properties. Hence, the considered datasets are:

4.1.1 KDDCup99 dataset [40]

This dataset was created in 1988 by the Defence Advanced Research Project Agency (DARPA) name Knowledge Discovery and Data Mining (KDD). The aim was to create a detailed and realistic IDS research benchmarking dataset with correctness and competence to inspect a real intrusive behavior. The dataset was collected through simulated intrusions on a modeled network in 2 months. It consists of a total of nearly 4,900,000 records of a series of TCP sessions. A portion of it with 2,000,000 connection records with 41 attributes was derived and used for the 3rd international KDD tools competition in 1999 and named the KDDCup99 dataset. It has 22 different classes of attacks for the target variable, and some of the classes belong to the minority.

4.1.2 NSL–KDD (Network Socket Layer–Knowledge Discovery in Databases) Dataset [41]

It has been derived from KDDCup99 by removing the redundancies [35]. This dataset is developed in two files—training and testing with 1,25,973 and 22,544 data samples and 41 attributes. The 22 types of attacks are grouped into four categories types of attacks. The target variable contains majority samples for normal, DoS attack, and the Probe attack classes and minority samples for R2L and U2R classes in the training file.

4.1.3 CICIDS-2017 dataset [42]

This dataset contains the realistic background of network traffic events generated by 25 users. The users' profiles were detailed, including specific purpose protocols such as SSH, HTTP, HTTPS, email, and FTP. Packet size distribution, packets per flow, payload size, and patterns in payload were demonstrated by utilizing the statistical metrics. This dataset covers different types of common attack scenarios. It is publically accessible in Packet CAPture (PCAP) format and CSV files format. There are eight CSV files, namely Friday-WorkingHours-Afternoon-DDos.csv, Friday-WorkingHours-Morning.csv, Friday-WorkingHours-Afternoon-PortScan.csv, Monday-WorkingHours.csv, Thursday-WorkingHours-Morning-WebAttacks.csv, Thursday-WorkingHours-Afternoon-Infiltration.csv, Tuesday-WorkingHours.csv, and Wednesday-workingHours.csv. The CICIDS-2017 was created with the actual traces of benign and attacks from the network traffic data, where 2,830,108 records are available in the dataset. 2,358,036 records belong to the benign traffic, and 471,454 records belong to attack records. The CICIDS-2017 contains 84 features, where the last column represents the traffic status or target attribute. This dataset is one of the popular datasets which includes up-to-date attacks as its uniqueness. Hence, the CICIDS-2017 has been considered the most relevant IDS benchmark dataset for evaluating the GWO–PSO–RF model. Table 2 depicts the distribution of fifteen class labels in the CICIDS-2017 dataset.

The non-homogeneous distribution of data samples in KDDCup99, NSL–KDD, and CICIDS-2017 datasets has motivated us to train the proposed NIDS model to identify both—majority and minority attacks.

4.2 Data preprocessing

All data from CSV files of CICIDS-2017 were collected into one data frame from the data collection module which is fed to the preprocessing module. The CICIDS-2017 dataset has been preprocessed using the preprocessing module to be parsed easily through the machine in ML processes. The

Table 2 Distribution of 15 class labels in the CICIDS-2017 dataset

Traffic type	Size	Traffic type	Size
Benign or Normal	2,358,036	DoS Slow HTTP Test	5499
DoS Hulk	231,073	Botnet	1966
Port Scan	158,930	Web Attack: Brute Force	1507
DDoS	41,835	Web Attack: XSS	625
DoSGoldenEye	10,293	Infiltration	36
FTP Patator	7938	Web Attack: SQL Injection	21
SSH Patator	5897	HeartBleed	11
DoS Slow Loris	5796		

same approach has also been repeated on the KDDCup99 and NSL–KDD datasets.

4.3 Feature selection (FS)

The preprocessed data have been applied to the feature selection module. The combination of GWO and PSO as hybrid GWO–PSO has been utilized to give a better feature subset selection.

4.3.1 Grey wolf optimization (GWO)

GWO is a metaheuristic search algorithm proposed by Mirjalili [43]. The grey wolves are arranging themselves at four levels ‘ α -wolves’ as first-level are leaders for making decisions, ‘ β -wolves’ as second-level for helping ‘ α -wolves’, ‘ δ -wolves’ as the third level to execute the instruction from ‘ α -wolves’ and ‘ β -wolves’, and ‘ δ -wolves’ as fourth-level acting like as an executor for all wolves. In this subsection, only the three main steps prey searching, prey encircling, and hunting will be discussed.

4.3.1.1 Prey searching To search the prey, random initializations of grey wolves within the searching space are generated, and they diverge from one to another in the search operation. When the prey is detected, the wolves are merged.

4.3.1.2 Prey encirclement On detection of the prey, the grey wolves start encirclement of prey, which can be presented mathematically as:

$$\vec{X}(t+1) = \vec{X}_p(t) - \vec{A} \cdot \vec{D}, \quad (1)$$

where $\vec{X}$ indicates grey wolf position, $\vec{X}_p$ indicates the prey position, t denotes the iteration number, and $\vec{D}$ can be calculated as:

$$\vec{D} = \left| \vec{C} \times \vec{X}_p(t) - \vec{X}(t) \right|, \quad (2)$$

$\vec{A}$ and $\vec{C}$ can be calculated as:

$$\vec{A} = 2\vec{a} \times \vec{r}_1 - \vec{a}, \quad (3)$$

$$\vec{C} = 2 \times \vec{r}_2, \quad (4)$$

where $\vec{a}$ decreased linearly from two to zero depending upon the iterations. The random vectors $\vec{r}_1$ and $\vec{r}_2$ are within the limit of [0, 1].

4.3.1.3 Hunting or chasing or optimization After the prey encirclement, the grey wolves focused on hunting the prey. The GWO hunting (optimization) is supervised by ‘ α ’, ‘ β ’,

and ‘ δ ’ wolves only. Aloha ‘ α ’ leads hunting while ‘ β ’ and ‘ δ ’ wolves participate in hunting. The following equations represent the positions of ‘ α ’, ‘ β ’, and ‘ δ ’ wolves:

$$\vec{D}_\alpha = \vec{C}_1 \times \vec{X}_\alpha - \vec{X}, \quad (5)$$

$$\vec{D}_\beta = \vec{C}_2 \times \vec{X}_\beta - \vec{X}, \quad (6)$$

$$\vec{D}_\delta = \vec{C}_3 \times \vec{X}_\delta - \vec{X}. \quad (7)$$

In addition, the positions of prey in the view of α , β , and δ wolves are represented by the following equations:

$$\vec{X}_1 = \vec{X}_\alpha - \vec{A}_1 \times \vec{D}_\alpha, \quad (8)$$

$$\vec{X}_2 = \vec{X}_\beta - \vec{A}_2 \times \vec{D}_\beta, \quad (9)$$

$$\vec{X}_3 = \vec{X}_\delta - \vec{A}_3 \times \vec{D}_\delta. \quad (10)$$

With the help of positions of α , β , and δ wolves, the best position of prey can be obtained as:

$$\vec{X}_t(t+1) = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{3} \quad (11)$$

4.3.2 Particle swarm Optimization (PSO)

Eberhart and Kennedy in 1995 proposed this PSO method as a population-based search method or algorithm. The population is updated through the fitness method’s values, which accepts the environment’s information to move the population towards a better solution space. Individuals or agents or particles are given velocity for flying in the searching environment. The experiences of particles and their acquaintances are used to adjust the momentum in a multidimensional search space [44]. Each particle has a new velocity in each iteration. A new position of them is updated accordingly, where the required parameters are considered with the present velocity of the particles. The distance of each particle from the best performance is obtained during the search process.

If $X_k^i = (X_{k1}^i, X_{k2}^i, \dots, X_{kD}^i)$ and $V_k^i = (V_{k1}^i, V_{k2}^i, \dots, V_{kD}^i)$ be the position and velocity of the i^{th} particle then maximum particle velocity (used to find out the searching regions between the current position and final position) and minimum velocity represented as:

$$V_i^{\max} = N \times (X_i^{\max} - X_i^{\min}); 0 < N \leq 1, \quad (12)$$

$$V_i^{\min} = -V_i^{\max}. \quad (13)$$

In addition, best position is given as:

$$Pb_i^k = (Pb_{i1}^k, Pb_{i2}^k, \dots, Pb_{id}^k). \quad (14)$$

If Gb^k is the indicator for achieving best performance achieved by the particle, then updates for velocity and position of each particle are as following:

$$V_i^{k+1} = \omega \times V_i^k + r_1 \times c_1 \times (Pb_i^k - X_i^k) + r_2 \times c_2 \times (Gb_i^k - X_i^k), \quad (15)$$

$$X_i^{k+1} = X_i^k + V_i^{k+1}, \quad (16)$$

where k is the iteration number, r_1 and r_2 are the random numbers between 0 and 1, c_1 and c_2 are positive constants used for defining weighting acceleration towards the 'Pb' and 'Gb' positions for each particle, and ' ω ' is considered as inertia weight.

4.3.3 Hybrid GWO–PSO

The drawbacks of the GWO, such as poor local minima searching ability, low accuracy, and slow convergence, are overcome using Gb^k best performance indicator value through PSO in the position update Eq. (11) of GWO. Therefore, this paper used a hybrid GWO–PSO algorithm to extract the best feasible features. The hybrid GWO–PSO and PSO algorithms are depicted in Figs. 3 and 4.

The considered objective function is the squared sum of values of each agent (attribute) within the loop. The objective function is provided below as:

$$F_1(x) = \sum_{i=0}^n x_i^2. \quad (17)$$

More other types of objective functions are mentioned in [45].

Algorithm 1 Hybrid GWO-PSO

Input: Dataset, Objective Function, number of iterations, population size
Output: best selected feature indexes
Initialize Positions from Dataset, score and position of α , β , and δ , search agents no., dimension.
Use PSO for receiving Gb^k as
Best=PSO(objective function, lb, ub, dim, population_size, max_iter)
Fitness=[]
n_features=Number of features to be selected
for ($l=0$; $l < \text{max_iter}$; $l++$) do
 Features={}
 for ($i=0$; $i < \text{dim}$; $i++$) do
 Fitness=Objective_Function(Current Agent)
 Update α _Score, α _Pos, β _Score, β _Pos, and δ _Score, δ _Pos on the basis of Fitness Value
 Feature[i]=Fitness
 end for
 Calculate $a = (2 - l * \frac{2}{\text{max_iter}})$
 for ($i=0$; $i < \text{search agent number}$; $i++$) do
 for ($j=0$; $j < \text{dim}$; $j++$) do
 Calculate The positions of α , β , and δ as D_α , D_β , D_δ and position of prey with respect α , β , and δ as $\vec{X}_1$, $\vec{X}_2$, $\vec{X}_3$ using the equations (5), (6), (7), (8), (9) and (10).
 Calculate new position of wolf as:
 $\vec{X}_{t+1} = \frac{\vec{X}_1 + \vec{X}_2 + \vec{X}_3}{\text{Best}}$
 end for
 end for
 Convergence_curve[l] = α _Score;
 Fitness.append(α _Score) as the relevant feature in an array
end for
return sorted Feature array

Fig. 3 Hybrid GWO–PSO algorithm

Algorithm 2 Particle swarm optimization (PSO)

Input: objective function, lb, ub, dim, population_size, max_iter

Output: G_best_score

Initialize max_velocity, Positions, P_best, P_best_score, G_best, G_best_score, velocity, wMax, wMin.

```

for (l=0; l < max_iter; l++) do
  for (i=0; i < population_size; i++) do
    Fitness=Objective_Function(Current Agent)
    Update P_best, P_best_score, and G_best, G_best_score on the basis of Fitness Value
  end for
  Calculate  $w = (wMax - l * \frac{wMax - wMin}{max\_iter})$ 
  for (i=0; i < population_size; i++) do
    for (j=0; j < dim; j++) do
      Calculate velocity using equation (15)
      Update position using equation (16).
    end for
  end for
  Convergence_curve[l] = G_best_score;
end for
return G_best_score

```

Fig. 4 PSO algorithm

Using hybrid GWO–PSO, all relevant features were extracted by ignoring irrelevant features, reducing the CICIDS-2017 dataset. Seventy-six feature's values have been applied as input to the hybrid GWO–PSO. The features names are given in Table 3.

Thirty features index was the output for which features name are provided in Table 4.

Similarly, on applying hybrid GWO–PSO, the indexes of selected features from KDDCup99 dataset are 4, 6, 8, 18, 19, 20, 21, 22, 23, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, and 37. In addition for NSL–KDD dataset are 1, 2, 3, 6, 7, 8, 9, 10, 11, 13, 14, 18, 20, 21, 22, 23, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 36, 37, 38 and 40.

4.4 Train test split and addressing of data unbalancing in training data

The collected data for selected features were divided into the training data samples and testing data samples. The training data samples were resampled so that the minority class data samples could get importance in the training phase and could be able to identify the attacks of minority class during the testing phase. Resampling techniques help to make the dataset granular for each class by adding or removing data samples. Hence, resampling affects class distribution. The balanced dataset is the transformed

dataset where the standard ML algorithms performed or fitted well. We have used an oversampling technique. This technique creates or repeats examples of minority classes.

4.5 Random forest classifier

Bagging classifier is a type of ensemble technique that is also known as bootstrap aggregation. Multiple base models ($M_1, M_2, \dots, M_n$) are combined. Each model is provided different samples of records using row sampling with replacement. In row sampling with replacement, some records may get repeated in the samples provided to the models. The voting classifier is used for aggregating the outputs of the models for reaching a decision. Random forest is a type of bagging classifier where many decision trees are used as multiple models. Each decision tree is given the input based on row sampling and column sampling. The decision tree has the problem that it has low bias and high variance. This means the tree has better performance on the training phase and poor performance on the testing phase. The voting scheme reduces the variance from high to low since the decision does not depend only on a particular tree, but it depends on the voting of multiple trees [46] (Fig. 5).

Hence, the IoT network's proposed IDS model is the combination of feature selection algorithm hybrid GWO–PSO and classification algorithm random forest (RF), which is represented as GWO–PSO–RF IoT IDS model.

Table 3 Feature set used for input to the hybrid GWO–PSO algorithm

S.I.	Attribute name	S.I.	Attribute name
1	Destination Port	39	Flow Duration
2	Total Fwd75Packets	40	Min Packet Length
3	Bwd IAT Std	41	Max Packet Length
4	Idle Min	42	Active Mean
5	Idle Max	43	min_seg_size_forward
6	Idle Std	44	act_data_pkt_fwd
7	Idle Mean	45	Init_Win_bytes_backward
8	Active Min	46	Init_Win_bytes_forward
9	Active Max	47	SubflowBwdBytes
10	Active Std	48	SubflowBwd Packets
11	BwdAvg Bytes/Bulk	49	'SubflowFwd Bytes
12	FwdAvg Packets/Bulk	50	SubflowFwd Packets
13	FwdAvg Bytes/Bulk	51	BwdAvg Bulk Rate
14	Fwd Header Length.1	52	BwdAvg Packets/Bulk
15	AvgBwd Segment Size	53	FwdAvg Bulk Rate
16	AvgFwd Segment Size	54	Average Packet Size
17	Down/Up Ratio	55	ECE Flag Count
18	URG Flag Count	56	CWE Flag Count
19	ACK Flag Count	57	PSH Flag Count
20	Bwd Packets/s	58	RST Flag Count
21	Fwd Packets/s	59	SYN Flag Count
22	BwdHeader Length	60	FIN Flag Count
23	Fwd Header Length	61	Packet Length Variance
24	Bwd URG Flags	62	Packet Length Std
25	Fwd URG Flags	63	Packet Length Mean
26	Bwd PSH Flags	64	Bwd IAT Min
27	Fwd PSH Flags	65	Bwd IAT Max
28	FwdIAT Mean	66	Bwd IAT Mean
29	Fwd IAT Total	67	Bwd IAT Total
30	Flow IAT Min	68	Fwd IAT Min
31	Flow IAT Max	69	Fwd IAT Max
32	Flow IAT Std	70	Flow IAT Mean
33	Bwd Packet Length Std	71	Fwd IAT Std
34	Fwd Packet Length Std	72	Bwd Packet Length Max
35	Fwd Packet Length Mean	73	Bwd Packet Length Min
36	Fwd Packet Length Min	74	Fwd Packet Length Max
37	Total Length of Bwd P76ackets	75	Bwd Packet Length Mean
38	Total Backward Packets	76	Total Length of Fwd Packets

5 Implementation and evaluation of results

This section discusses the experimental setup, considered metrics for evaluating the model, evaluation, results, and different comparisons of the proposed NIDS model with others' works. At last, the discussion is presented.

Table 4 Selected features as the output of the hybrid GWO–PSO algorithm

S.I.	Attribute name	S.I.	Attribute name
1	PSH Flag Count	16	Init_Win_bytes_forward
2	Bwd IAT Std	17	Fwd IAT Std
3	Flow IAT Std	18	Average Packet Size
4	Fwd IAT Mean	19	Packet Length Mean
5	FIN Flag Count	20	Init_Win_bytes_backward
6	Bwd Packet Length Std	21	Bwd Packet Length Mean
7	Destination Port	22	Bwd Packet Length Max
8	Max Packet Length	23	ACK Flag Count
9	Packet Length Std	24	Idle Min
10	AvgBwd Segment Size	25	Idle Mean
11	Bwd IAT Max	26	Fwd IAT Max
12	Flow IAT Max	27	URG Flag Count
13	Idle Max	28	Bwd IAT Total
14	SYN Flag Count	29	Fwd IAT Total
15	Fwd PSH Flags	30	Flow Duration

5.1 Experimental setup

The proposed model's performance assessment has been carried out on an HP machine operated on Intel Core i7-5500U CPU at 2.40 GHz 2401 MHz, 2 Cores 4 Logical processors with 12 GB RAM running Microsoft Windows 10 Professional. The feature selection algorithms and classifiers are implemented in Python programming language (Version 3.7.3). To set up the experimental environment, Anaconda Navigator has been installed on the mentioned machine.

5.2 Performance evaluation metrics

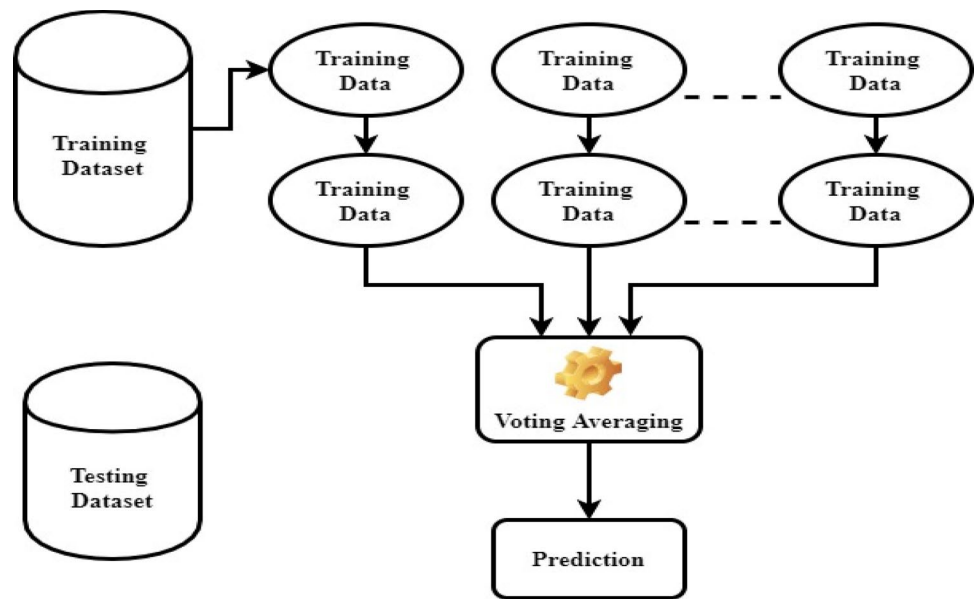
Accuracy, Precision, recall, or attack detection rate (DR), and F1-score are considered for the performance analysis of the proposed GWO–PSO–RF IDS model for IoT network which are discussed [47] as following:

5.2.1 Accuracy

It is the ration of true predicted classes and total actual classes and is represented by

$$\text{FNR} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}},$$

where TP is the true Benign network prediction, and TN is true attack network prediction, and FN is the false attack network prediction for Benign networks. FP is a false Benign network prediction for attack networks.

Fig. 5 Random Forest classifier

5.2.2 Precision

Indicate the ratio of positively predicted values that are being positive. Higher Precision shows low FPR:

$$\text{Precision} = \frac{TP}{TP + FP}.$$

5.2.3 Recall or attack detection rate (ADR)

It defines the capability of detecting various attacks in the IoT networks:

$$\text{Recall} = \frac{TP}{TP + FN}.$$

5.2.4 F1-score

The F1-score of a model is calculated based on precision rate and recall and is represented as:

$$\text{F1 Score} = \frac{2 \times (\text{Recall} \times \text{Precision})}{(\text{Recall} + \text{Precision})}.$$

For conducting the comprehensive performance evaluation of various ML algorithms, the experiments have been carried out through the train test split validation method on the considered datasets. The method has divided each dataset into two parts, like 75% and 25%, to create a training test and a testing set.

Table 5 Classification report of GWO–PSO–RF model on the test set

Target class	Precision	Recall	F1-score	Support
Benign or Normal	1.00	1.00	1.00	568,719
DoS Hulk	0.85	0.80	0.82	488
Port Scan	1.00	1.00	1.00	31,818
DDoS	1.00	0.99	1.00	2603
DoSGoldenEye	1.00	1.00	1.00	57,680
FTP Patator	0.99	0.99	0.99	1353
SSH Patator	0.99	0.99	0.99	1479
DoS Slow Loris	1.00	1.00	1.00	1945
DoS Slow HTTP Test	1.00	1.00	1.00	2
Botnet	1.00	0.60	0.75	10
Web Attack: Brute Force	0.99	1.00	1.00	39,515
Web Attack: XSS	1.00	1.00	1.00	1474
Infiltration	0.71	0.76	0.74	404
Web Attack: SQL Injection	1.00	0.43	0.60	7
HeartBleed	0.50	0.31	0.35	189
Weighted avg	1.00	1.00	1.00	707,686

5.3 Evaluation of results

For evaluating the results of the proposed GWO–PSO–RF IoT network IDS model, we compared the results with the other classifiers, namely decision tree and logistic regression. The results are obtained from the unbalanced test set of each dataset. After that, the proposed GWO–PSO–RF IDS model results on test sets of different datasets have been compared with other similar approaches.

Confusion Matrix :

[[567358	247	68	53	172	91	75	10	3	10
298	5	197	1	131]					
[18	470	0	0	0	0	0	0	0	0
0	0	0	0	0]					
[7	0	31810	0	0	1	0	0	0	0
0	0	0	0	0]					
[7	0	0	2590	4	1	0	0	0	0
0	0	0	1	0]					
[4	0	1	7	57658	0	0	0	0	0
9	0	1	0	0]					
[1	0	0	1	0	1343	8	0	0	0
0	0	0	0	0]					
[5	0	0	2	2	6	1462	0	0	0
1	0	1	0	0]					
[0	0	0	0	0	0	0	1945	0	0
0	0	0	0	0]					
[0	0	0	0	0	0	0	0	2	0
0	0	0	0	0]					
[0	0	0	0	0	0	0	0	0	10
0	0	0	0	0]					
[1	0	0	0	4	0	0	0	0	0
39507	0	3	0	0]					
[0	0	0	0	0	0	0	0	0	0
0	1474	0	0	0]					
[1	0	0	0	0	0	0	0	0	0
1	0	300	9	93]					
[0	0	0	2	2	0	0	0	0	0
0	0	0	3	0]					
[2	0	0	0	0	2	1	0	0	0
1	0	114	0	69]]					

Fig. 6 Confusion matrix for the test set of CICIDS-2017

5.3.1 Evaluation of CICIDS-2017 dataset

Table 5 shows the values of considered metrics to each class for the GWO–PSO–RF IoT network IDS model on the test set of the CICIDS-2017 dataset.

Figure 6 depicts the confusion matrix of the proposed GWO–PSO–RF IoT network IDS model on the unbalanced test set of the CICIDS-2017 dataset for multiclass classification.

Table 6 and Fig. 7 represent the detection rate (DR) comparison of the proposed model from other recent approaches. The observation shows that the DR of the balanced GWO–PSO–RF model for each class is higher than the different recent methods. It has been observed from Table 5 that the model predictions on the unbalanced test set are poor only for Infiltration, and HeartBleed. However,

comparatively detection rate for the other 12 classes is better than the different approaches presented in Table 6.

Table 7 and Fig. 8 show the accuracy comparison of the proposed NIDS model on the CICIDS-2017 dataset with other approaches.

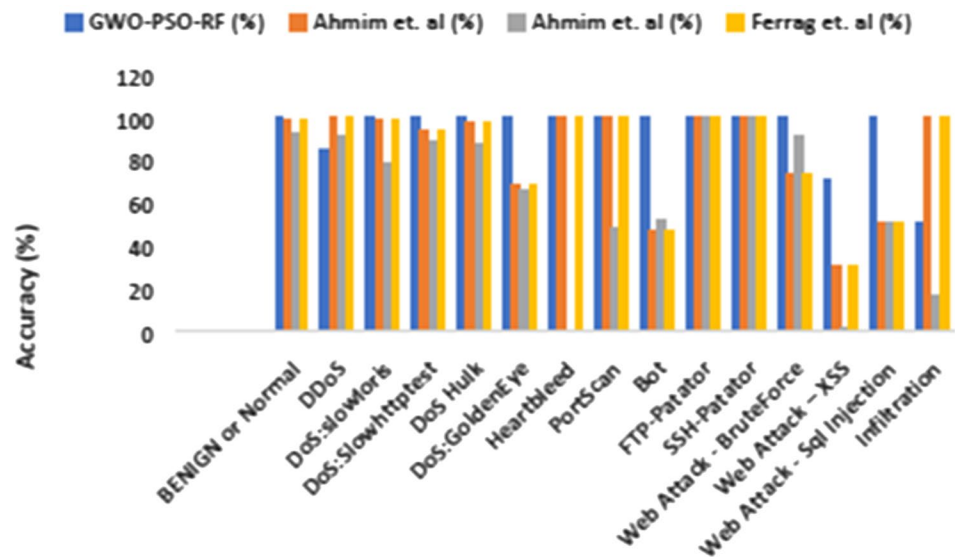
5.3.2 Evaluation of NSL–KDD dataset

Figure 9 depicts the confusion matrix, accuracy, and other considered metrics values on the test set of the NSL–KDD dataset after applying the proposed NIDS model for multiclass classification.

The comparison achieved accuracy from the proposed GWO–PSO–RF NIDS model and recent models on the NSL–KDD dataset are depicted in Table 8 and Fig. 10.

Table 6 DR comparison of GWO_PSO_FS_RF with other approaches on the class basis

Target class	GWO_PSO_FS_RF (%)	Ahmim et. al. [1] (%)	Ahmim et. al. [1] (%)	Ferrag et. al. [2] (%)
BENIGN or normal	100	98.86	92.65	98.86
DDoS	85	99.88	91.21	99.88
DoS:slowloris	100	97.76	78.49	97.76
DoS:Slowhttptest	100	93.84	88.54	93.84
DoS Hulk	100	96.78	86.89	96.78
DoS:GoldenEye	99	67.57	65.43	67.57
Heartbleed	99	100.00	0.00	100
PortScan	100	99.88	48.52	99.88
Bot	100	46.47	51.28	46.47
FTP-Patator	100	99.64	99.00	99.64
SSH-Patator	99	99.91	99.73	99.91
Web attack—BruteForce	100	73.27	90.41	73.27
Web attack—XSS	71	30.63	1.88	30.63
Web attack—Sql injection	100	50.00	50.00	50.00
Infiltration	50	100.00	16.67	100.00

Fig. 7 DR comparison of the proposed model from other recent approaches**Table 7** Comparison of accuracy of the proposed model with other approaches on CICIDS-2017 dataset

Model	Accuracy
Adaboost [51]	81.83
Deep Neural Network [52]	96.3
Ensemble Classifier [53]	96.8
GRU-RNN [54]	89
k-Neighbors [55]	94
Proposed (GWO-PSO-RF)	99.88

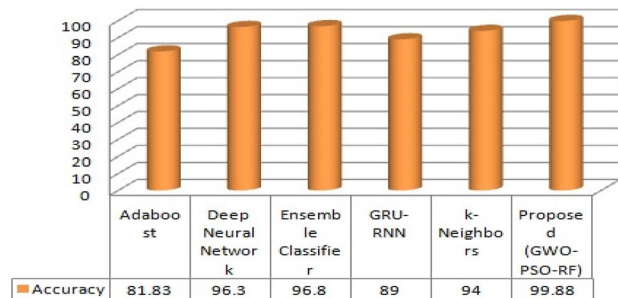
**Fig. 8** Comparison of accuracy of the proposed model with other approaches on CICIDS-2017 dataset

Fig. 9 Confusion matrix, accuracy, and other metrics on the test set of NSL–KDD dataset after applying the proposed NIDS model for multiclass classification

Confusion Matrix :					
[	13433	1	0	6	8]
[	0	56	3	2	3]
[	1	6	880	4	15]
[	0	0	1	3494	9]
[	23	5	145	52	18982]]
Accuracy Score : 0.9923509924856581					
Report :					
		precision	recall	f1-score	support
0	1.00	1.00	1.00	1.00	13448
1	0.82	0.88	0.85	0.85	64
2	0.86	0.97	0.91	0.91	906
3	0.98	1.00	0.99	0.99	3504
4	1.00	0.99	0.99	0.99	19207
accuracy				0.99	37129
macro avg	0.93	0.97	0.95	0.95	37129
weighted avg	0.99	0.99	0.99	0.99	37129

Table 8 Accuracy comparison of proposed GWO–PSO–RF NIDS model from works of others on NSL–KDD dataset

References	Accuracy
ANN [56]	78.51
Vinayakumar et al. [52]	80.10
Diro et al. (2018) [57]	99.20
Li et al. [58]	94.92
Yanmiao et al. [59]	86.95
Proposed (GWO–PSO–RF)	99.24



Fig. 10 Accuracy comparison of proposed GWO–PSO–RF NIDS model from works of others on NSL–KDD dataset

5.3.3 Evaluation of KDDCup99 dataset

Figure 11 depicts the values of considered metrics values on the test set of the KDDCup99 dataset after applying the proposed NIDS model for multiclass classification.

The comparison achieved accuracy from the proposed GWO–PSO–RF NIDS model and recent models on the NSL–KDD dataset are depicted in Table 9 and Fig. 12.

Figure 13 depicts the performances of Random Forest, Decision Tree, Logistic Regression, and Naive Bayesian classifiers with GWO–PSO on KDDCup99, NSL–KDD, and CICIDS-2017 datasets.

Figure 14 depicts the accuracies of GWO–PSO–RF on KDDCup99, NSL–KDD, and CICIDS-2017 datasets, for different situations when the number of decision trees varies within the random forest for voting the prediction for the intrusions.

Figure 15 depicts the accuracies of GWO–PSO–RF on KDDCup99, NSL–KDD, and CICIDS-2017 datasets, for different situations when test data sizes are being changed.

5.4 Discussion

In this paper, supervised ML-based classifiers are considered. In the supervised ML approach, the target values are known to the classifier during the training phase. The results of different considered classifiers and the comparison of proposed one from others' works show the effectiveness of supervised ML approaches in intrusion detection purposes. The main reason for choosing the supervised ML approach is that the network traffic pattern can be learned effectively during the training phase to predict intrusions or attacks efficiently. These characteristics or patterns can be categorized between Benign and attack using different network-based features. The other ML approaches can also be used to perform similar tasks. The other ML approaches can be any unsupervised, semi-supervised, or reinforcement learning. Clustering can help to improve the learning of different classifiers. In this paper, the emphasis is mainly on the supervised ML approach, where the input data have been made balanced for training the model. The model has been tested

Fig. 11 Values of considered metrics on applying the proposed NIDS model for multiclass classification on the test set of KDDCup99 dataset

	precision	recall	f1-score	support
0	1.00	1.00	1.00	537
1	0.20	0.33	0.25	6
2	0.50	1.00	0.67	1
3	0.94	0.94	0.94	17
4	0.75	1.00	0.86	3
5	0.98	0.99	0.99	315
6	1.00	1.00	1.00	3
7	0.00	0.00	0.00	4
8	0.00	0.00	0.00	0
9	1.00	1.00	1.00	26865
10	0.92	0.87	0.89	53
11	1.00	1.00	1.00	24392
12	0.00	0.00	0.00	0
13	1.00	1.00	1.00	1
14	0.99	0.97	0.98	72
15	1.00	1.00	1.00	275
16	0.00	0.00	0.00	5
17	0.98	0.99	0.99	393
18	1.00	1.00	1.00	70070
19	0.00	0.00	0.00	1
20	1.00	1.00	1.00	253
21	0.93	0.97	0.95	234
22	0.17	0.60	0.26	5
accuracy			1.00	123505
macro avg	0.67	0.72	0.69	123505
weighted avg	1.00	1.00	1.00	123505

Table 9 Accuracy comparison of proposed GWO-PSO-RF NIDS model from works of others on KDDCup99 dataset

Model	Accuracy
ANN [56]	92.39
LSTM-RNN [60]	96.93
BRNN [60]	99.4
GRU-RNN [61]	97.06
All Layers IDS [62]	98.91
Proposed (GWO-PSO-RF)	99.66

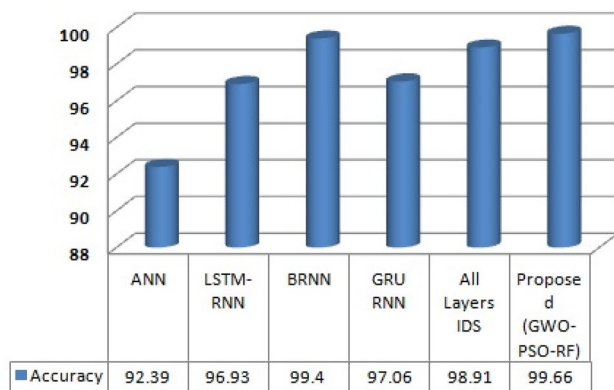


Fig. 12 Accuracy comparison of proposed GWO-PSO-RF NIDS model from works of others on KDDCup99 dataset

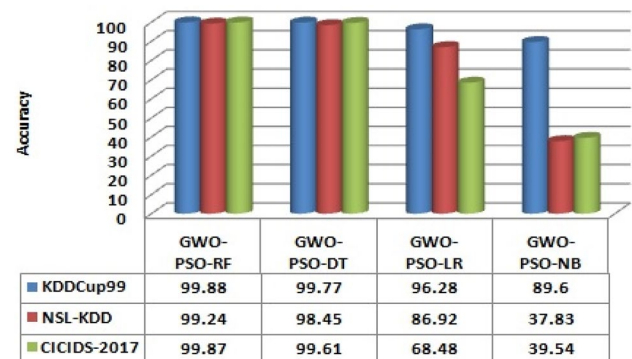


Fig. 13 Accuracy of designed and developed NIDS models on the datasets - KDDCup'99, NSL-KDD, and CICIDS-2017

on an unbalanced test set. For a balanced test set, it has been observed that the proposed GWO-PSO-RF NIDS model for the IoT network has achieved up to 100% accuracy for each class in the case of binary and multi-classification. Since the model is not getting balanced data for the real environment classification; hence, the proposed model has not been evaluated on balanced test data that has not been presented in this paper. A population-based metaheuristic approach as a hybrid GWO-PSO algorithm has been applied for selecting relevant features. Therefore, that the further processing, memory requirement, and time complexity can be reduced to fit the proposed model in the IoT environment. The proposed

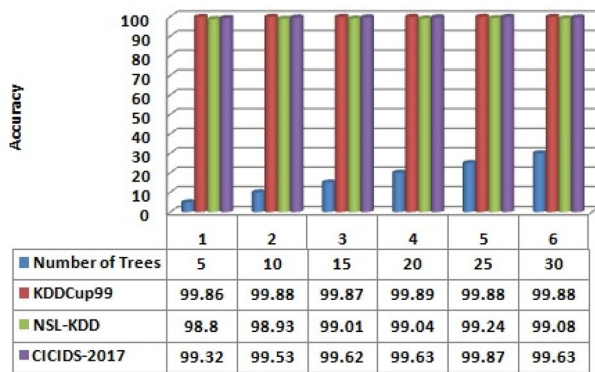


Fig. 14 Accuracy of proposed NIDS model on changing the number of decision trees in RF for KDDCup'99, NSL-KDD, and CICIDS-2017 datasets

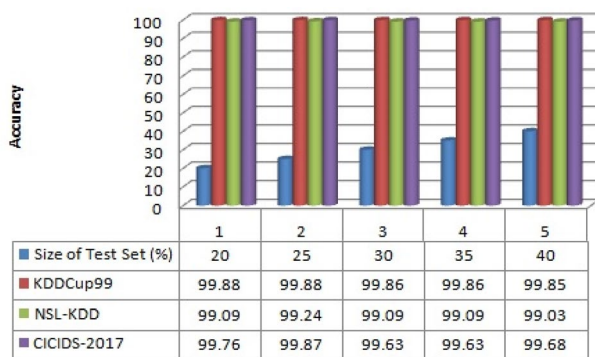


Fig. 15 Accuracy of proposed NIDS model on changing the training and testing size within KDDCup'99, NSL-KDD, and CICIDS-2017 datasets

model achieved maximum accuracy for the training size of 25% of the whole dataset. The number of estimators (trees) was 25, and the criterion was entropy in the random forest classifier. In the future, we can implement the knowledge distillation technique to train the proposed NIDS model with some other ML approaches to fit the model in the small IoT devices.

6 Conclusion and future direction of work

In this paper, a network intrusion detection framework and a model have been proposed for the IoT network environment. The model was implemented in the python programming language. The model was evaluated KDDCup99, NSL-KDD, and CICIDS-2017 datasets using accuracy, Precision, recall, and F1-score. The dataset is available in eight CSV files combined to start training a balanced

GWO-PSO-RF IDS model for the IoT network. The balanced GWO-PSO-RF NIDS model was evaluated on 25% of different datasets. The results were compared with the decision tree (DT), logistic regression (LR), and Naive Bayesian (NB) algorithms with hybrid GWO-PSO. It has been shown that balanced data training removes minority class prediction problem while unbalanced dataset training overlooked the minority classes on the prediction. The proposed model has also been compared with other recent approaches and achieved the highest accuracy of 99.66% on average for all considered datasets. Thus, the proposed balanced GWO-PSO-RF IDS model for IoT network addressing the biasing problem towards the more frequently occurring records and increases the DR for minority classes of attacks.

Our future direction of the work is to implement the proposed model in a real online learning environment for intrusion detection for protecting the IoT network environment effectively in an intelligent manner. In the future, the distillation technique can be utilized to design the NIDS to be placed within the IoT device.

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